DATA ITEM DESCRIPTION

Title: **ACQUISITION AND SUSTAINMENT DATA PACKAGE (ASDP) PRODUCT ENGINEERING DESIGN DATA AND ASSOCIATED LISTS (EDDAL)**

Number: DI-SESS-82351
Approval Date: 20210415
AMSC Number: F10247
Limitation: N/A
DTIC Applicable: No
GIDEP Applicable: No
Preparing Activity: 11 (AFLCMC/EZSC)
Project Number: SESS-2021-002
Applicable Forms: N/A

Use/Relationship: The Acquisition and Sustainment Data Package (ASDP) Product Engineering Design Data and Associated Lists (EDDAL) provide engineering data adequate to serve as an authoritative and complete technical description of an item. This data, when authorized, is adequate to support competitive procurement and maintenance for items, including for substitutability, backward compatibility, and interchangeability with the as-designed and as-delivered assets. The ASDP Product EDDAL shall provide the design, engineering, manufacturing, and sustainment information to procure or manufacture an item to be identical to the original item. The product shall be defined to the extent necessary for a competent manufacturer to produce an item that duplicates the physical, interface, and functional characteristics of the original product, as-delivered, without additional processing, design engineering effort, or recourse to the design activity. This data represents the highest level of design disclosure.

a. This Data Item Description (DID) is applicable to acquisitions of military systems, equipment, and components, including spares. It is intended to be used for acquiring Product EDDALs at the end of the Engineering and Manufacturing Development Phase and during subsequent phases of the Department of Defense (DoD) materiel life cycle to define the approved as-designed configuration (product baseline).

b. A purchased item, as defined by American Society of Mechanical Engineers (ASME) Y14.24, *Types and Applications of Engineering Drawings, Engineering Drawing and Related Documentation Practices*, is an item that is sold or traded in the course of conducting normal business operations, is used by commercial industry, or is a specialized version of a supplier's general product line that is routinely customized. (Copies of this document are available online at www.asme.org.) Purchased items as used herein have also been referred to as vendor items or vendor-developed items.

c. This DID contains the format, content, and intended use information for the data deliverable resulting from the work task described in 5.3 and 5.4.1.3 of MIL-STD-31000, *Technical Data Packages*, as related to ASDP Product EDDAL. (Copies of this document are available online at https://quicksearch.dla.mil.)

d. This DID is related to DI-SESS-81000, *Product Engineering Design Data and Associated Lists*, but is created to meet specific enterprise Product Life Cycle Management (PLM) business requirements. Other related design data DIDs include: DI-SESS-81001, *Conceptual Design Drawings/Models*, DI-SESS-81002, *Developmental Design Drawings/Models and Associated Lists*, and DI-SESS-81003, *Commercial Engineering Design Data and Associated Lists*. (Copies of these documents are available online at https://quicksearch.dla.mil.)

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Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The ASDP Product EDDAL shall be in the native format and in accordance with (IAW) ASME Y14.100, *Engineering Drawing Practices* and Appendices B through F within; ASME Y14.24; ASME Y14.34, *Associated Lists*; ASME Y14.35, *Revision of Engineering Drawings and Associated Documents*; ASME Y14.41, *Digital Product Definition Data Practices*; ASME Y14.47, *Model Organization Practices*; and ASME Y14.5, *Dimensioning and Tolerancing*. (Copies of these documents are available online at www.asme.org.) In addition, the ASDP Product EDDAL formatting shall be IAW MIL-STD-31000. The ASDP Product EDDAL shall be in the English language. The final ASDP Product EDDAL shall have all outstanding changes incorporated into the data. Printed Wiring Boards (PWBs) shall be written in American Standard Code for Information Interchange (ASCII) and comply with current Gerber RS-274X Format Users Guide. (Copies of this document are available online at www.ucamco.com/en/.)

2.1 The Data Objects/Attributes and Associated Metadata (DOAM) listing shall be in an Excel populated spreadsheet format IAW the template provided in the DOAM Specification attached to the contract. Each artifact and data set submitted IAW this DID shall have a designated row in the template.

3. Content.

3.1 General. The ASDP Product EDDAL shall meet the requirements of MIL-STD-31000 and describe the complete physical and performance characteristics of an item or component in sufficient detail to ensure that an item or component produced IAW this data is identical to the original item or component. The engineering design data shall:

a. Reflect the end product at its current level of design maturity.

b. Provide the engineering data for logistics support products.

c. Provide the data necessary to permit competitive acquisition, repair, and manufacture of the original item(s).

d. Provide the required tying, linking, and tracing component, subassembly, and assembly information to produce the lowest level design and specification for a digital representation of a physical item (digital thread of the end item).

e. Provide details of any unique processes and specifications (not published or available to industry) utilized to build the end items IAW the contract.

3.2 The ASDP Product EDDAL shall conform to the requirements of MIL-STD-31000; ASME Y14.100 and Appendices B through F within; ASME Y14.24; ASME Y14.34; ASME Y14.35; ASME Y14.41; ASME Y14.47; and ASME Y14.5. Metadata for each artifact and dataset shall be IAW ASME Y14.47 and the DOAM Specification. (Copies of ASME documents are available online at www.asme.org.) Product Engineering Design Data of end items, elements, and all components, assemblies, and related special test equipment and interface adaptors shall be provided to define the details necessary for the manufacture, test, inspection, maintenance, operations, and logistic support of the system. The ASDP Product EDDAL shall specify and reflect, directly or by reference, the following:

a. Details of unique processes and specifications (i.e., not published or generally available to

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industry) when essential to design and manufacture.

b. Performance ratings.

c. All dimensional and tolerance data needed to fully define the part or assembly, including geometric dimensioning and tolerancing (GD&T) where necessary to define and control dependencies and critical interfaces.

d. Manufacturing processes, specifications, and assembly sequences.

(1) If a manufacturing process or specification is required, it shall be clearly annotated in the engineering design data. Any process or specification referenced within the engineering data that is a non-public or non-military standard shall be defined within the applicable engineering design data.

e. Tolerance input and output characteristics.

f. Diagrams with vector starting points and dimensional scaling.

g. Mechanical and electrical connections.

h. Physical characteristics, including interface(s), form, finishes, and protective coatings.

i. Details of material characteristics, heat treatments, conditions, coatings and plating, surface finishes and treatments, and all other data necessary to define the material condition for all components and their locations in the assembly.

j. Inspection, test, and evaluation criteria. First Article Acceptance, Lot test plans and procedures, Acceptance Test Procedures (ATP), Non-Destructive Inspections (NDI), and any Military screening shall be referenced to verify proper performance.

k. Quality assurance requirements for test, inspection, acceptance and verification information. Engineering design data shall reference the information required to test, inspect, accept, and verify quality assurance requirements to the level specified in the contract or purchase order. All Quality Assurance Provisions (QAP) shall be clearly annotated identifying any special tests, inspections, measurements, or certifications required to ensure the item being defined meets its intended performance.

(1) All Critical Safety Items (CSI) and Critical Safety Processes (CSP) shall be clearly annotated in the engineering design data. Definition and conforming markings shall be IAW ASME Y14.100.

(2) All Safety Critical Functions (SCF) and Safety Critical Items (SCI) shall be clearly annotated in the engineering design data with a pointer to the component that is deemed SCI or SCF. Definitions of SCI and SCF shall be IAW MIL-STD-882, *System Safety*, and MIL-HDBK-516, *Airworthiness Certification Criteria*. (Copies of these documents are available online at https://quicksearch.dla.mil.)

(3) All Mission Critical Function or Mission Critical Item (MCF or MCI, respectively) shall be clearly annotated in the engineering design data. Definitions of MCF and MCI shall be IAW MIL-HDBK-516, under “Mission Critical”. “MCF” or “MCI” shall be clearly annotated in the engineering design data with a pointer to the component(s) that are deemed MCF or MCI.

l. Hardware marking requirements. Engineering design data shall identify the unique identifier (UID) and unique item identifier (UII) marking requirements and methods for identification to be utilized and applied, and shall provide the exact location of the marking and the information to be contained within the marking, per MIL-STD-130, *Identification Marking of U.S. Military Property.* (Copies of this document are available online at

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https://quicksearch.dla.mil.) For material subject to the marking requirements of other specifications or standards, the engineering design data shall specify the marking method, location, and content of those markings. Engineering design data containing statements for items, such as: "Government Furnished Property" or "Government Furnished Equipment" shall also include the contract number providing the government furnished property or equipment to the contractor (for example: "Government Furnished Property under F34601-99-#-####.").

m. References to requirements for reliability, maintainability, environmental conditioning, shock and vibration testing, and other operational or functional tests.

n. Vendor substantiated data.

o. References to requirements for programming software into devices or assemblies, including a description of the input media and the procedures for validating that the software has been installed correctly.

p. Interconnection diagrams required for wiring harnesses and cable assemblies.

q. Special Test Equipment (STE) and Special Inspection Equipment (SIE) (to include special test fixtures).

(1) Equipment calibration requirements shall be identified.

r. Warranty Statements. Warranty statements shall be in the engineering design data. The statements shall not obliterate, nor obscure, information on the field of the engineering design data. The warranty statement shall specify the warranty period, all actions that would render it as voided, and include the contractor or agency that shall correct or repair the deficiency. Warranty statements in the engineering design data shall contain the following information:

(1) "Warranted item until (warranty expiration date using YYMMDD format)."

(2) Contract number (i.e., the contract number that invokes the warranty of the engineering design data).

(3) "Deficiencies in the engineering design data shall be corrected by (enter the contractor and contractor's address or the agency that shall correct the deficiency)."

3.3 Item definition. All parameters required to define each unit, assembly, subassembly, part or component, or material shall be included in the ASDP Product EDDAL. This shall include the following data:

a. All necessary mechanical dimensions to fully describe fabrication, acceptance, interface, or installation of the item defined.

b. All necessary electrical parameters to fully describe fabrication, acceptance, interface, or installation of the item defined.

c. All other necessary physical parameters to fully describe fabrication, part and material location(s), acceptance, interface, and installation of the item defined (i.e., weight, pressure, viscosity, etc.).

d. All necessary environmental conditions that units, assemblies, subassemblies, parts, and materials must meet to perform effectively in the end item, such that the end item will meet its specification requirements.

e. Cable Interconnect Diagrams (CIDs); electrical system schematics; cable harness assembly models and drawings; PWB assemblies; printed wiring layer, artwork, and wiring lists; and fluid system schematics, which shall be IAW ASME Y14.100. The engineering

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design data shall include the following:

(1) The CID shall show graphically the arrangement of external electrical cabling that interconnects electrical assemblies and equipment. The CID shall show all cable runs and terminations and each cable shall be identified by a reference designation number.

(2) Electrical system schematics shall illustrate and describe circuit items with symbols placed such that a circuit can be traced from item to item in the sequence of its function. The placement and arrangement of the circuits shall follow a logical sequence of presentation to provide a clear description of the distribution of power.

(3) Cable harness assembly models or drawings shall meet the requirements of the Drawing Requirements Manual (DRM) and ASME Y14.24. (Copies of the DRM are available online at https://ihsmarkit.com/index.html.)

(4) Component Level Documentation. Schematics and wiring lists for components, including interconnecting cable harnesses, shall be provided.

(5) Overall Grounding Documentation. The grounding schematics shall show the details of all grounds and power returns from source to loads. All connections shall be shown. It shall also show details of all electrical ground support equipment (EGSE) interconnections to facility and safety grounds. The grounding schematics shall meet the requirements of the DRM.

(6) The fluid system schematics shall illustrate and describe all components with symbols and flow designators such that the fluid system can be traced from component to component (such as: pumps, valves, meters, regulators, and filters). The schematics shall document the range requirements (i.e., flow, temperature, and pressure) for all component external interfaces and line sizes. The placement and arrangement of the components shall follow a logical sequence of presentation to provide a clear description of the flow of fluids in the system.

(7) PWB description shall be IAW American National Standards Institute (ANSI)/IPC-D-350, *Printed Board Description in Digital Form*; IPC-2221, *Generic Standard on Printed Board Design*; and IPC-2611, *Generic Requirements for Electronic Product Documentation*. (Copies of these documents are available online at https://www.ipc.org.) The PWB shall include the following:

(a) Delivery media shall have a CONTENTS.TXT file in ASCII format listing files on the delivery media, drawing or model number, part number(s), assigned Commercial and Government Entity (CAGE) code(s), data rights, size of each file in uncompressed form, application data, revision level of the file, and the purpose of the file.

(b) Delivery media containing more than one PWB assembly shall contain directories for each PWB.

(c) Comply with the current Gerber RS-274X Format Users Guide. (Copies of this document are available online at www.ucamco.com/en/).

(d) Provide aperture wheel table or a list of correlating apertures with "D" codes (D-codes are instructions to the photoplotter that naturally include the letter "D."). If the table or list is contained in a separate document, identify the document in a file and supply the document with the PWB. Identify the type and model of plotter data that is intended to be used on.

(e) Identify computer aided design (CAD) platform and software used to develop Gerber file.

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(f) Identify each dataset such that each dataset can be identified by using drawings and models. Board data sets shall include the following:

1 Revision letter of aperture file (if applicable).

2 Identify board part number.

3 CAGE code.

4 Column headings shall identify horizontal (width) and vertical (height) dimensions which correspond to “X” and “Y” coordinates, respectively, for proper orientation of shapes.

5 Units of measurement (e.g., mils, millimeters, etc.).

(g) The board files shall be properly aligned and include the following images:

1 Circuit layers.

2 Board outlines.

3 Silk screen of component reference designations.

4 Silk screen of component’s outlines as they appear on the Printed Circuit Board (PCB).

5 Solder mask.

6 Plated and un-plated drill holes.

(h) Each of the images shall include the following:

1 Board part number.

2 Revision letter of image.

3 CAGE Code.

4 Layer order number and image name.

5 Vertical and horizontal register marks with x-y coordinate dimensions and tolerances.

(i) The drill data shall contain the following information:

1 Board part number.

2 Revision.

3 CAGE Code.

4 File type (e.g., Excellon, Plain ASCII Text, etc.).

5 Drill tool sizes.

6 Holes x-y coordinates.

7 Drill tool (bit) used at each x-y location.

8 Identification of plated and un-plated holes, if not identified on the master drawing or model.

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3.4 The ASDP Product EDDAL shall depict the as-designed Bill of Material (BoM) (i.e., Engineering Bill of Material (EBoM)) for each approved product configuration, graphically depict the hierarchical structure of all parts from the top-level configuration item down to assemblies, subassemblies, and components with reference to the associated engineering design data.

3.5 CAGE code and document numbers. The ASDP Product EDDAL shall be identified with the contractor's CAGE code and contractor document numbers or with a Government CAGE code and document numbers as specified in the contract.

3.6 Vendor Item Control Documents. Vendor Item Control Documents shall be used to specify the requirements for purchased items when such items have been approved, by the Government, for use in the design and they are used without alteration, selection, or source qualification (testing of an item prior to procurement action to ensure that it satisfies the specified requirements). Vendor Item Control Drawings, when selected in lieu of, or in addition to, control drawings for repairable commercial items, shall be included and provide information to permit Government maintenance and modification of the items. Control drawings shall disclose engineering form, fit, function, and performance requirements for the acquisition of interchangeable purchased items of existing designs, or of items specially developed by vendors to the control drawing requirements.

3.7 Source Control Documents. Source Control Documents shall be included to specify the requirements for purchased items only when such items have been approved by the Government for use in the design and:

a. The item is for a critical application,

b. The requirements can be met by an item from one or more sources, and

c. The application requires source qualification (testing of a critical item prior to procurement action to ensure that it satisfies the specified requirements).

d. Shall meet ASME Y14.24.

3.7.1 Form, fit, and function requirements conforming to the requirements for a control drawing as defined in ASME Y14.100 and ASME Y14.24 shall be specified for engineering data for which the Government does not have unlimited rights.

3.8 Standard Microcircuit Drawings and Models shall be IAW MIL-HDBK-780, *Standard Microcircuit Drawings*. (Copies of this document are available online at https://quicksearch.dla.mil.)

3.9 Associated List. All application lists, data lists, index lists, parts lists, and wire lists shall meet the requirements of ASME Y14.34. All associated lists needed to describe the complete physical and performance characteristics of an item or component being acquired shall be contained in the ASDP Product EDDAL.

3.10 Distribution Statement. A Distribution Statement (i.e., A, B, C, D, E, etc.) IAW Department of Defense Instruction (DoDI) 5230.24, *Distribution Statements on Technical Documents*, Enclosure 4, shall be added. (Copies of this document are available online at www.esd.whs.mil/DD.)

3.11 Classification Marking. The applicable classification marking, as necessary, in accordance with Department of Defense Manual (DoDM) 5200.01, Volume 2, *DoD Information Security Program: Marking of Information*, shall be added. (Copies of this document are available online at www.esd.whs.mil/DD.)

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3.12 DOAM**.** The completed ASDP Product EDDAL template, provided in the DOAM Specification attached to the contract, shall comply with the format requirements above so that it can be ingestible into a PLM solution to ensure proper tying, tracing, and linking. Any field that is not applicable shall be marked “NA.”

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